

Week 17 Progress Report – VerifAI

Overview

This week focused on extending the VerifAI framework from unimodal deepfake detection to a multimodal audio-visual pipeline and integrating the trained model into a working API for inference.

Multimodal Model Implementation

A late-fusion audio-visual model was implemented using the FakeAVCeleb v1.2 dataset, which contains aligned video and audio streams.

- Visual branch: MobileNetV2 backbone processing 5 uniformly sampled frames per video.
- Audio branch: ResNet18-based encoder processing MFCC features.
- Fusion strategy: Concatenation of audio and visual embeddings followed by an MLP classifier.

The model was trained for 5 epochs on the FakeAVCeleb train split and validated on the validation split.

Training Results

The fusion model achieved strong performance:

- Validation Accuracy $\approx 99.8\%$
- F1-score ≈ 0.999
- AUC ≈ 0.999

Embedding norm logging confirmed that both audio and visual branches contributed non-trivial signals, validating correct multimodal fusion.

Evaluation

Evaluation on the FakeAVCeleb test split produced:

- Accuracy: 0.9995
- F1-score: 0.9997
- AUC: 0.9999

Only a single misclassification was observed, indicating near-perfect performance on the in-distribution dataset.

API Integration

A FastAPI-based inference backend was implemented to satisfy the Week 17 requirement of beginning system integration.

The API provides:

- A /api/predict endpoint accepting uploaded video files.
- On-the-fly preprocessing:
 - Extraction of 5 video frames.
 - Audio extraction and MFCC computation.
- Inference using the trained multimodal fusion model.
- JSON response containing:
 - Predicted label (real/fake).
 - Probability of being fake.

End-to-end testing via Swagger UI confirmed that:

- The backend runs successfully.
- Video uploads are processed correctly.
- Model inference is executed, and results are returned.

Observations & Limitations

While the API functions correctly, predictions on arbitrary “in-the-wild” videos may differ from expectations. This is due to:

- Lightweight inference-time preprocessing
- Domain shift between FakeAVCeleb and real-world deepfake videos
- Absence of face alignment and dataset-specific normalisation in the API pipeline

These limitations are expected at this stage and do not affect the validity of the Week 17 objectives.